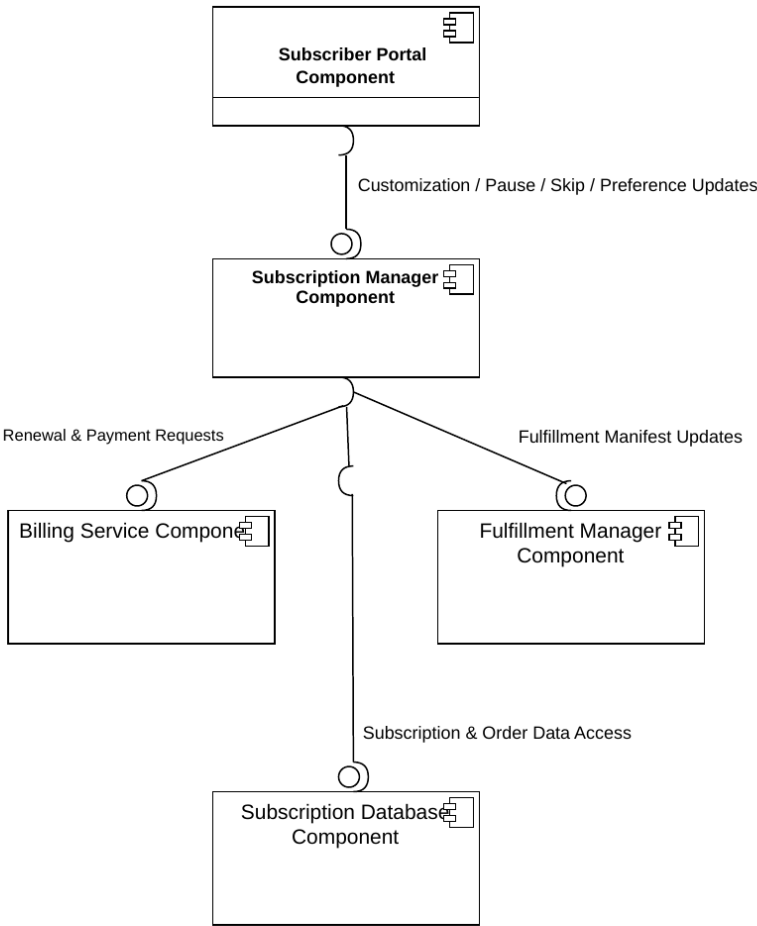


Customizable Subscription Box Scheduler

Lab 3 - Component Diagram

Customizable
Subscription Box
Scheduler
Component Diagram

Name:AVANEESH
SRN:PES1UG24CS099
Section:B



Customizable Subscription Box Scheduler

Architectural Justification

Name	AVANEESH
SRN	PES1UG24CS099
Section	B
Primary Domain	Retail, E-Commerce & Finance
Target Actors	Subscriber, Fulfillment Lead

Architecture Selection

We chose **Microservices Architecture** for the Customizable Subscription Box Scheduler. The system is divided into focused components for subscriber interaction, subscription management, billing, fulfillment, and subscription data, allowing the major functions to remain independently organized.

Two Reasons for the Choice

- 1. Independent scaling and deployment:** Subscription management, billing, and fulfillment have different workloads. Microservices allow these functions to be developed, deployed, and scaled independently as demand changes.
- 2. Fault isolation and modularity:** A problem in one service, such as fulfillment processing, can be isolated from other functions such as subscriber customization or billing. This separation also makes individual components easier to modify or replace.

Security Advantage

Security responsibilities can be isolated by service. In particular, billing/payment-related processing can be separated from subscriber-facing functionality, reducing the amount of the system that directly handles sensitive payment operations and allowing access controls to be applied at service boundaries.

Performance Benefit

The architecture supports independent scaling of high-load functions. For example, fulfillment manifest generation can be scaled separately from the subscriber portal during monthly processing peaks, while the subscriber-facing service remains responsive for customization requests.

Component Alignment

Component	Primary responsibility
Subscriber Portal Component	Subscriber customization, pause, skip, and preference interactions
Subscription Manager Component	Subscription business logic and coordination
Billing Service Component	Renewal and payment processing requests
Fulfillment Manager Component	Fulfillment manifest updates and shipping preparation
Subscription Database Component	Subscription and order data persistence and access

Prepared for Lab 3 - Component Modelling & Architectural Pattern Selection

Diagram and justification are combined here as a single submission PDF.